

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$ (8.1).

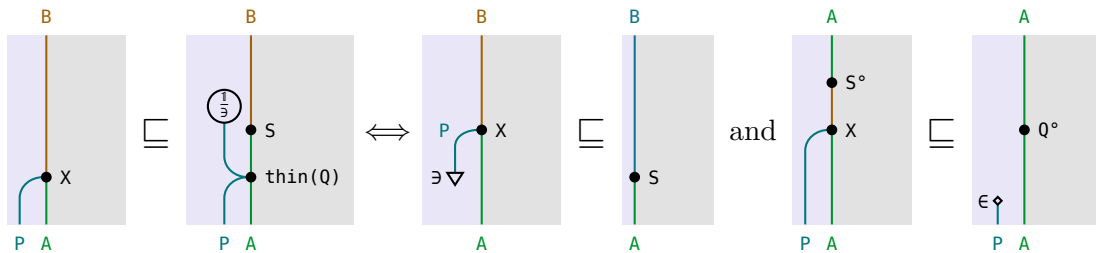
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

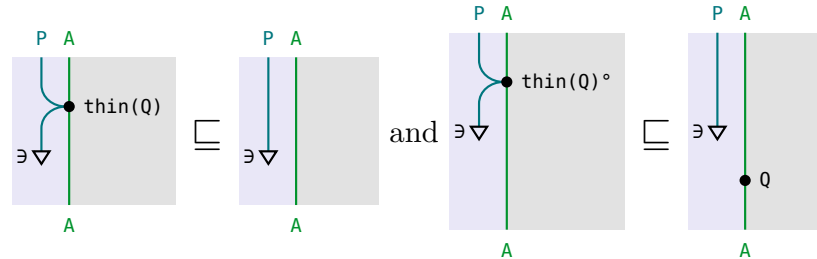
the law	what it says
1 $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ \exists \ \subseteq \ \exists$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ \exists \ \subseteq \ \exists \ Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \ Q \subseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \subseteq R \wedge 1 \subseteq Q \wedge R \ R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \supseteq \text{est}(Q) \ \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)

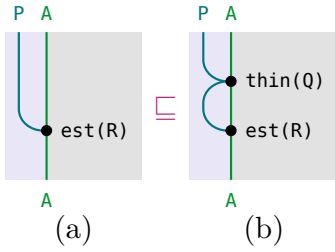


(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

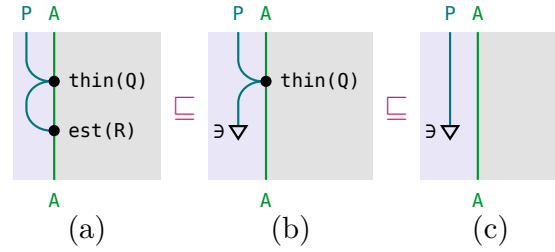
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$



(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — (8.1a.5)}$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

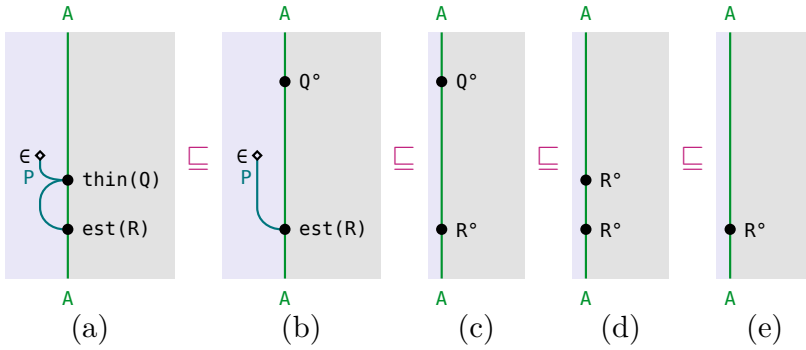
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — (7.1.1a)



(a)	$\text{thin}(Q) \text{ est}(R)$	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge R R \sqsubseteq R \Rightarrow \epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive



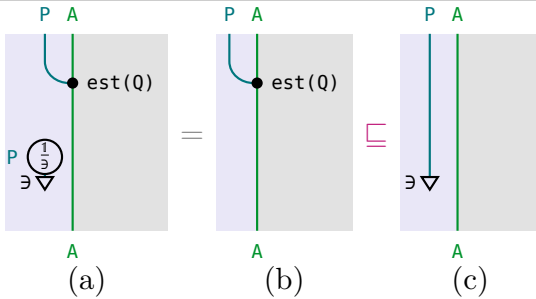
(a)	$\epsilon \text{ thin}(Q) \text{ est}(R)$	$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$
(b)	$\sqsubseteq Q^\circ \epsilon \text{ est}(R)$	$\epsilon \text{ est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	$R R \sqsubseteq R$
(e)	$\sqsubseteq R^\circ$	$\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$

$$\text{est}(Q) \frac{1}{3} \sqsubseteq \text{thin}(Q) \quad (8.2)$$

the singleton holding a Q -least member of a set is a thinning of that set

$$\text{est}(Q) \frac{1}{3} \exists \sqsubseteq \exists$$

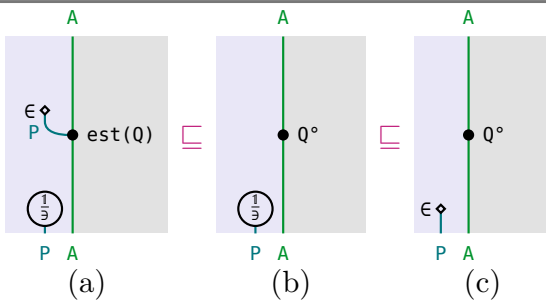
every member of the singleton is a member of the set — $\triangleq \text{thin}, \exists/\exists$ half



(a)	$\text{est}(Q) \frac{1}{3} \exists$	$\frac{1}{3} \exists = 1$
(b)	$= \text{est}(Q)$	$R \cap S \sqsubseteq R$
(c)	$\sqsubseteq \exists$	

$$\epsilon \text{ est}(Q) \frac{1}{3} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in Q to every member of the set — $\triangleq \text{thin}, \epsilon \backslash (Q^\circ \epsilon)$ half



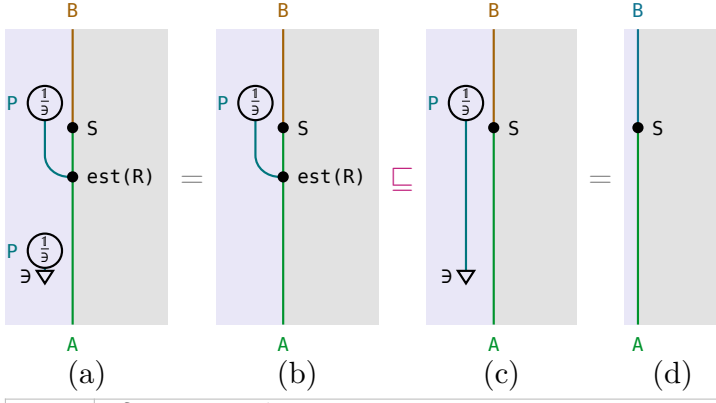
(a)	$\epsilon \text{ est}(Q) \frac{1}{3}$	$\epsilon \text{ est}(R) \sqsubseteq R^\circ$
(b)	$\sqsubseteq Q^\circ \frac{1}{3}$	$\frac{1}{3} \sqsubseteq \epsilon$
(c)	$\sqsubseteq Q^\circ \epsilon$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

given $R \cap (S^\circ S) \subseteq Q$, Q a preorder

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \subseteq S$$

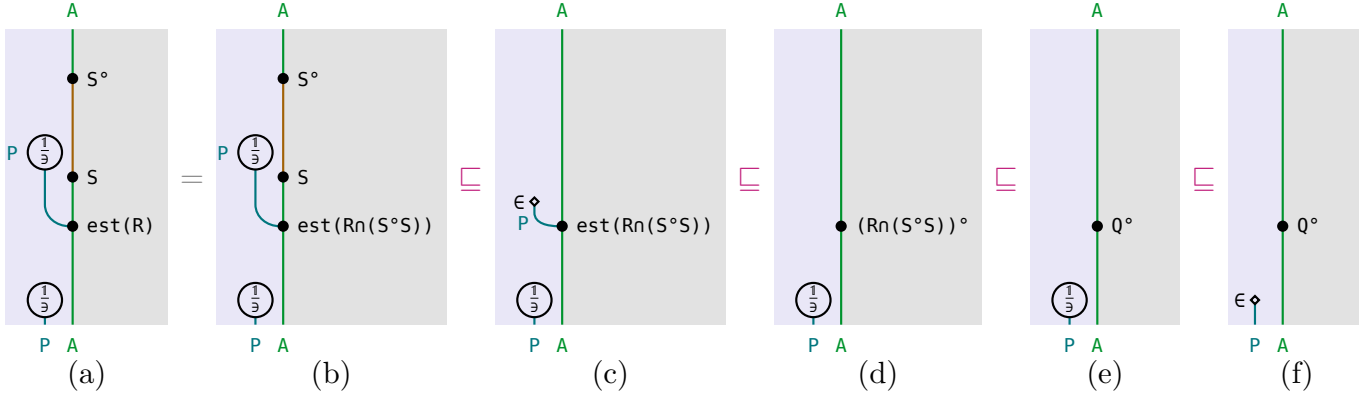
every member of the singleton is a value S returns — (8.1a.1), first condition



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$	$\frac{1}{\exists} \ni = 1$
(b)	$= \frac{S}{\exists} \text{est}(R)$	$R \cap S \subseteq R$
(c)	$\sqsubseteq \frac{S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — (8.1a.1), second condition



(a)	$S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$
(b)	$= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$T^\circ \frac{1}{\exists} \subseteq \in$
(c)	$\sqsubseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$\in \text{est}(R) \subseteq R^\circ$
(d)	$\sqsubseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$	$R \cap (S^\circ S) \subseteq Q$
(e)	$\sqsubseteq Q^\circ \frac{1}{\exists}$	$\frac{1}{\exists} \subseteq \in$
(f)	$\sqsubseteq Q^\circ \in$	

$$Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

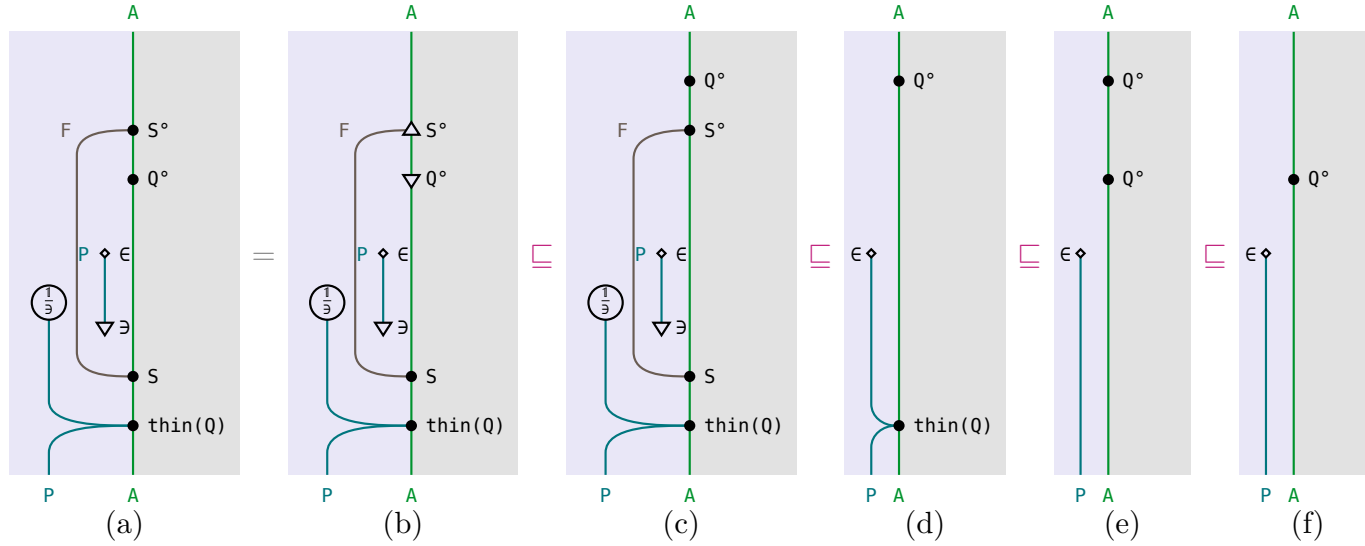
Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{ and } \langle \frac{(S)}{\exists} \rangle \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \in$$

(8.1a.1) at $X \triangleq \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle$, $\langle \frac{(S)}{\exists} \rangle$ for its S

$\Rightarrow$ fusion, hylomorphism theorem

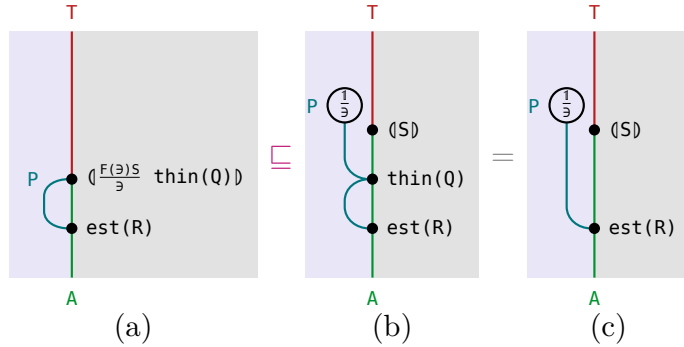


(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$t(f \ g) = t(f) \ t(g)$
(b)	$= S^\circ F(Q^\circ) F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\text{MonoAlg}(S, Q)$ $\Rightarrow S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
(c)	$\sqsubseteq Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(\in) \frac{F(\exists) S}{\exists} \sqsubseteq \in$
(d)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(e)	$\sqsubseteq Q^\circ Q^\circ \in$	$R \ R \sqsubseteq R$ $\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$
(f)	$\sqsubseteq Q^\circ \in$	

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders



(a)	$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	Theorem 8.1
(b)	$\sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \sqsubseteq R$ $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$
(c)	$= \langle \frac{(S)}{\exists} \text{est}(R) \rangle$	

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head} \ \text{head}^\circ)$,

$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl} \ P(r)]$,

$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} \ P(l), \text{cpr} \ P(r)]$, $\text{step} \triangleq \text{cpr} \ P(\text{cons}) \ \text{est}(R)$.

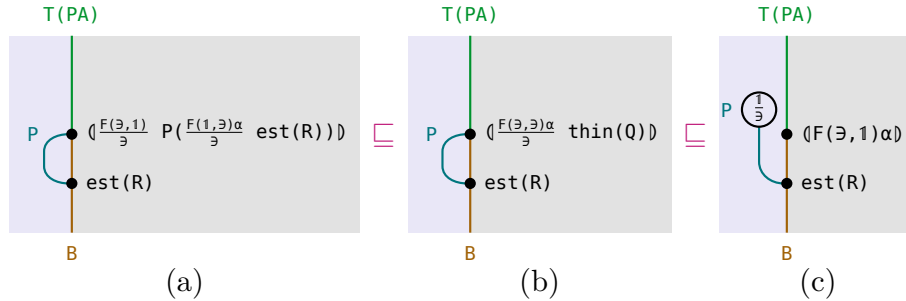
(8.2b)

the law	what it says
1 $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2 $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \cap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \sqsubseteq Q \quad (8.2c)$$

$$\Rightarrow (F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R)) \text{ est}(R) \sqsubseteq (F(\exists, 1) \alpha) \text{ est}(R)$$

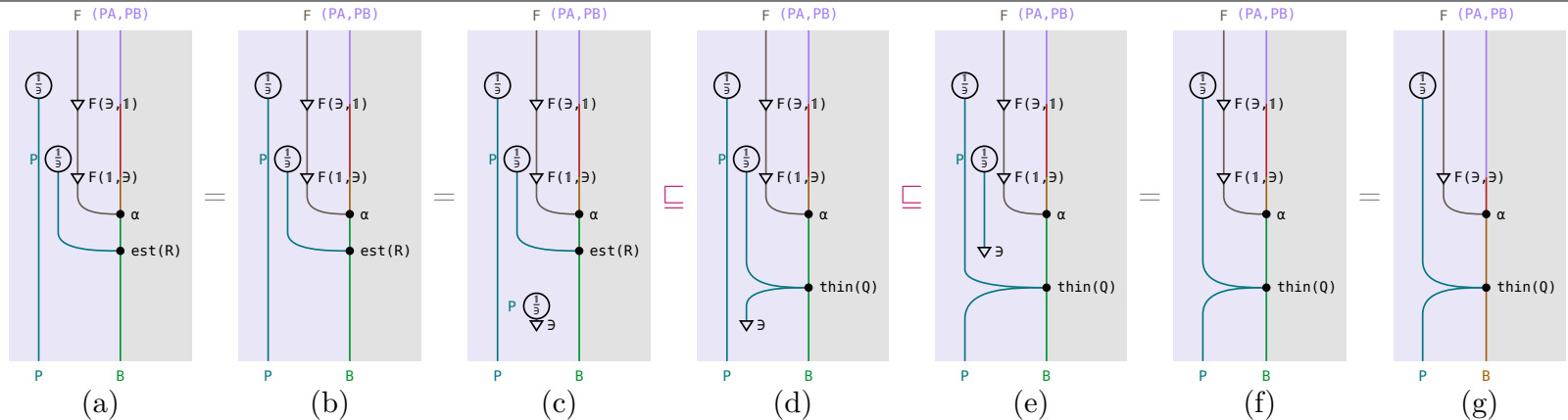
a least-cost path in a layered network, as a fold over the layers



(a)	$(F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R)) \text{ est}(R)$	(8.2d)
(b)	$\sqsubseteq (F(\exists, \exists) \alpha) \text{ thin}(Q) \text{ est}(R)$	$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q)$
(c)	$\sqsubseteq (F(\exists, 1) \alpha) \text{ est}(R)$	$\Rightarrow (F(\exists, \exists) \alpha) \text{ thin}(Q) \text{ est}(R) \sqsubseteq (F(\exists, 1) \alpha) \text{ est}(R)$

$$S \triangleq F(1, \exists) \alpha, R \cap (S^\circ S) \sqsubseteq Q \Rightarrow (F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R)) \sqsubseteq (F(\exists, \exists) \alpha) \text{ thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$(F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R))$	
(b)	$= (F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R))$	$P(\frac{1}{\exists}) E(\exists) = 1$
(c)	$= (F(\exists, 1) \alpha) P(F(1, \exists) \alpha \text{ est}(R) \frac{1}{\exists}) E(\exists)$	$P(R S) = P(R) P(S)$
(d)	$\sqsubseteq (F(\exists, 1) \alpha) P(F(1, \exists) \alpha) P(\text{thin}(Q)) E(\exists)$	(8.3)
(e)	$\sqsubseteq (F(\exists, 1) \alpha) P(F(1, \exists) \alpha) E(\exists) \text{ thin}(Q)$	$P(R S) = P(R) P(S)$
(f)	$= (F(\exists, 1) \alpha) F(1, \exists) \alpha \text{ thin}(Q)$	$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$
(g)	$= (F(\exists, \exists) \alpha) \text{ thin}(Q)$	$\frac{V S}{\exists} = \frac{V}{\exists} P(\frac{S}{\exists}) \text{ union}$
		$F(R, 1) F(1, S) = F(R, S)$

$$= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): (F(1, \exists) \alpha) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \quad | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and $d R d'$: the leaf moves by R;
- u, v are both inr and $p_1 R q_1 \wedge p_2 S q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr: $R + S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1) = 1$ allows no crossing.

(8.2e)

- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p)$, $\text{inr}(a, q)$, $\text{inr}(b, p)$, $\text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a)$, $\text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow PA$, $\text{cup} : PA \times PA \rightarrow PA$, $\text{cp}(F)$ (5.6a.12), $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$ for $\leq$ a connected preorder.

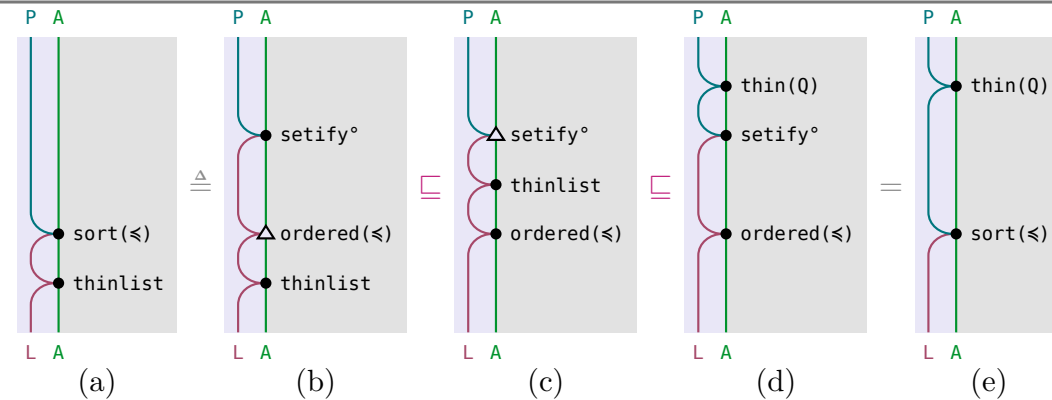
$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $Q[\text{nil}, \text{bump } Q]$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \ Q \ a \rightarrow [a] \# xs, a \ Q \ b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(8.3b)

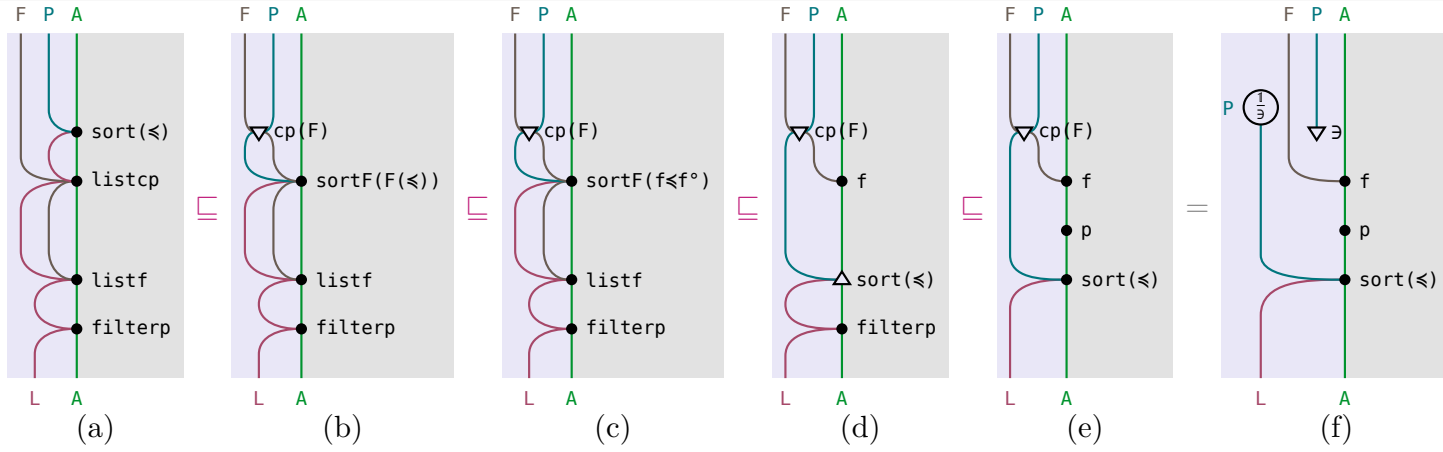
the law	what it says
1 $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
2 $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3 $\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
4 $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
5 $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
6 $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered($\leq$)) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq (8.6)
ordered($\leq$) $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sort($\leq$) thinlist $\sqsubseteq$ thin(Q) sort($\leq$)
 a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



(a)	$\text{sort}(\leq) \text{ thinlist}$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$
(b)	$\triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$	$\text{Coreflexive}(\text{ordered}(\leq)) \wedge g \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq} \text{ ordered}(\leq) \Rightarrow \text{setify}^\circ \text{ ordered}(\leq) g \sqsubseteq \text{setify}^\circ g \text{ ordered}(\leq)$
(c)	$\sqsubseteq \text{setify}^\circ \text{ thinlist} \text{ ordered}(\leq)$	$\text{Map}(\text{setify}) \wedge g \text{ setify} \sqsubseteq \text{setify } T \Rightarrow \text{setify}^\circ g \text{ ordered}(\leq) \sqsubseteq T \text{ setify}^\circ \text{ ordered}(\leq)$
(d)	$\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$	
(e)	$= \text{thin}(Q) \text{ sort}(\leq)$	

$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, \leq) \wedge \text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq \text{Lemma 8.1}$
 $\text{P}(f) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp} \sqsubseteq \text{E}(p) \text{ sort}(\leq) \wedge \text{F}(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(\text{F}) \text{ sortF}(\text{F}(\leq))$
 $\Rightarrow \text{F}(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp} \sqsubseteq \frac{\text{F}(\exists) \text{ f p}}{\exists} \text{ sort}(\leq)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : \mathbf{FA} \rightarrow \mathbf{A}$ monotonic
 on $\leq$, p coreflexive, F linear.



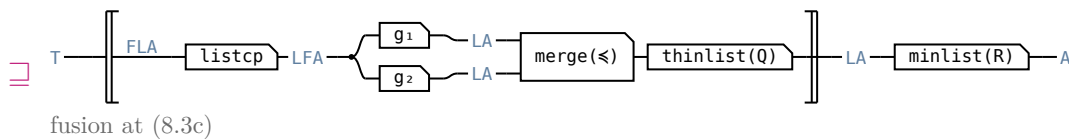
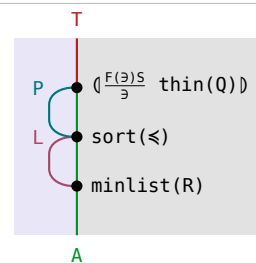
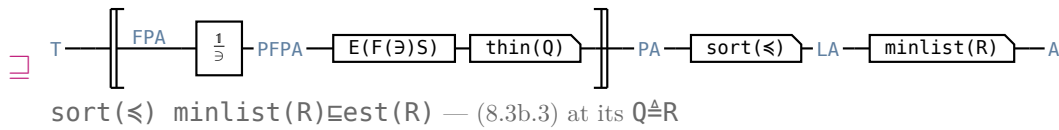
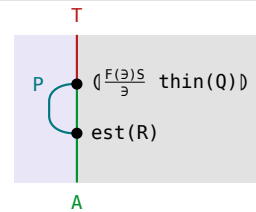
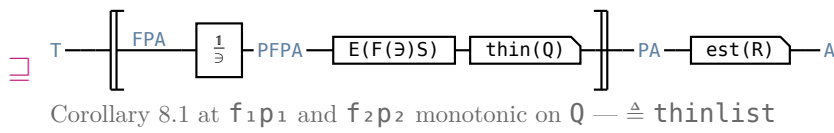
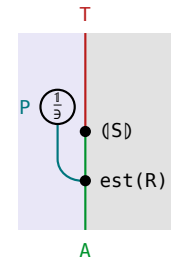
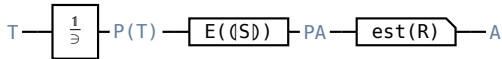
(a)	$\text{F}(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp}$	$\text{F}(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(\text{F}) \text{ sortF}(\text{F}(\leq))$
(b)	$\sqsubseteq \text{cp}(\text{F}) \text{ sortF}(\text{F}(\leq)) \text{ listf} \text{ filterp}$	$\text{MonoAlg}(f, \leq)$
(c)	$\sqsubseteq \text{cp}(\text{F}) \text{ sortF}(f \leq f^\circ) \text{ listf} \text{ filterp}$	$\text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq \text{P}(f) \text{ sort}(\leq)$
(d)	$\sqsubseteq \text{cp}(\text{F}) \text{ P}(f) \text{ sort}(\leq) \text{ filterp}$	$\text{sort}(\leq) \text{ filterp} \sqsubseteq \text{E}(p) \text{ sort}(\leq)$ $\text{Map}(f)$ $\Rightarrow \text{P}(f) = \text{E}(f)$ $\text{E}(R \ S) = \text{E}(R) \ \text{E}(S)$
(e)	$\sqsubseteq \text{cp}(\text{F}) \ \text{E}(f \ p) \ \text{sort}(\leq)$	$\frac{S}{\exists} \ \text{E}(R) = \frac{S \ R}{\exists}$
(f)	$= \frac{\text{F}(\exists) \text{ f p}}{\exists} \text{ sort}(\leq)$	

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1 \ p_1, Q) \wedge \text{Theorem 8.2}$
 $\text{MonoAlg}(f_2 \ p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq \text{P}(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$
 $\text{P}(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq \text{E}(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq \text{E}(p_2) \text{ sort}(\leq)$
 $\wedge \text{F}(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(\text{F}) \text{ sortF}(\text{F}(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = (f_1 \ p_1) \cup (f_2 \ p_2) \Rightarrow (\text{listcp} \ (g_1, g_2) \text{ merge}(\leq))$
 $\text{thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{\text{QS}}{\exists} \text{ est}(R)$

a fold on sorted lists of partial solutions, thinned at every step — at thinlist 's binary thinning data.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
$\sqsubseteq$ $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \text{ — (8.3b.2)}$	
$=$ $\langle \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \rangle \text{ cup} \text{ sort}(\leq) \text{ thinlist}(Q)$ $S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}$	
$\sqsubseteq$ $\langle \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q)$ $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \text{ — (8.3b.6)}$	
$=$ $\langle \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \rangle \text{ merge}(\leq) \text{ thinlist}(Q)$ $\langle X, Y \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) = \langle X \text{ sort}(\leq), Y \text{ sort}(\leq) \rangle \text{ — absorption}$	
$\sqsubseteq F(\text{sort}(\leq)) \text{ listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq) \text{ thinlist}(Q)$ Lemma 8.1 at f_1, p_1 and at f_2, p_2, then $X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — (5.2.2a.4); } g_i \triangleq \text{list}(f_i) \text{ filter}(p_i) \text{ — } \triangleq \text{thinlist}$	

§8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{ value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{ weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \circ \text{ , ,}$ within w the coreflexive on xs with weight $xs \leq w, 0 \leq w.$

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times A, \text{ listcp} = \text{wrap} + \text{cpr}, g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1],$
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

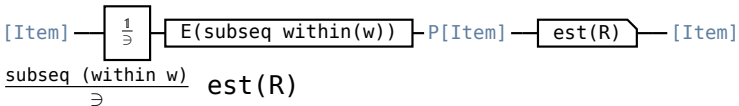
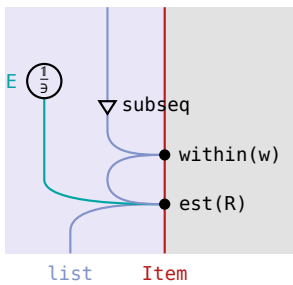
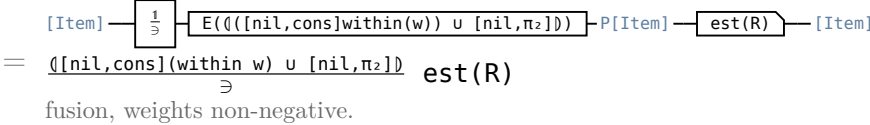
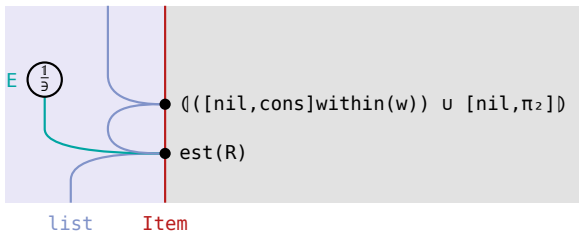
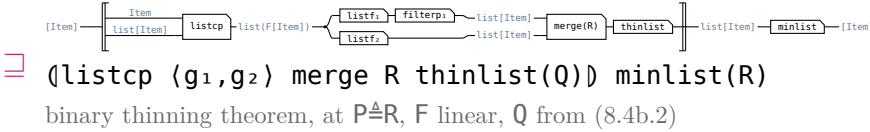
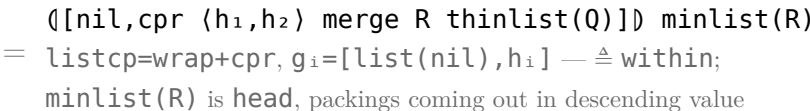
$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), h_2 \triangleq \text{list}(\pi_2).$

(8.4b)

the law	what it says
1 $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
2 $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R \\
& [\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq \\
& P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup } \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \\
& \text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow \langle \text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist} \rangle \\
& \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)
\end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
	
	
	
	

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

	the law	what it says
1	(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2	(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°,	both halves are monotonic on Q once ties in waste are broken by the first line
3	merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

($\forall a. 0 \leq \text{len}(a) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$ (8.5c)
 $\text{wrap}, \text{new}] \sqsubseteq [\text{wrap wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap wrap}, \text{glue}] \sqsubseteq \tau$
 $[\text{wrap wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}(w)) \text{sort}(\tau)$
 $\wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup} \text{sort}(\tau) \wedge$
 $\text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}$
 $(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\tau) \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\text{est}(R)}$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits}(w))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{\text{est}(R)}$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\text{est}(R)}$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\text{est}(R)}$ the algebra as $(f_1 p_1) \sqcup (f_2 p_2)$, $p_1 \triangleq 1 \triangleq \text{thinlist}$.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\sqsubseteq ([\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ binary thinning theorem, at $P \triangleq T$ with merge $\tau = \text{cat}$ ((8.5b.3)), Q from (8.5b.2)	
$([\text{start}, \text{cpr} (h_1, h_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct } \triangleq \text{partition}$	

§8.6. Bitonic tours

(8.6a)

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start} (a, b) = ([a, b], [a, b])$, $\text{dropl} (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr} (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}])$.

$\text{cost} (xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost } [a_0, \dots, a_n] = \text{tc} (a_0, a_1) + \dots + \text{tc} (a_{n-1}, a_n)$, $\text{incost } [a_0, \dots, a_n] = \text{tc} (a_1, a_0) + \dots + \text{tc} (a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $Q \triangleq R \cap (\text{next2 next2}^\circ) \cap (\text{head2 head2}^\circ)$, $P \triangleq T$, $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

(8.6b)

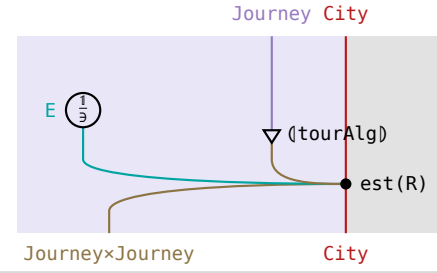
the law	what it says
1 $(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE $(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
2 $(1 \times Q) \text{dropl} \sqsubseteq \text{dropl } Q$ $(1 \times Q) \text{dropr} \sqsubseteq \text{dropr } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq \quad (8.6c) \\
& P([\text{start}, \text{droplFn}]) \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq \\
& P([\text{start}, \text{droprFn}]) \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \\
& \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp} \\
& (g_1, g_2) \text{cat thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{est}(R) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

circuit

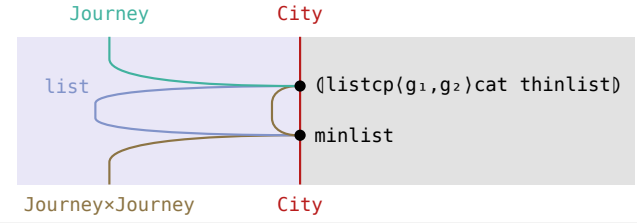
Hinze–Marsden

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& \frac{\text{tour}}{\exists} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& = \frac{([\text{start}, \text{dropl} \cup \text{dropr}])}{\exists} \text{est}(R) \\
& \text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}]) \triangleq \text{tour}
\end{aligned}$$

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& \sqsubseteq (\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)) \text{minlist}(R) \\
& \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from (8.6b.2)}
\end{aligned}$$



$$\begin{aligned}
& ([\text{start wrap}, \text{cpr} (\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& = \text{thinlist}(Q)] \text{minlist}(R) \\
& \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$